

$$x = 324$$

$$\log_x 27 - \log_{x^2} 81 = 1$$

$$\frac{\log_x 27 - \log_{x^2} 81}{\log_x x^2} = 1$$

$$\frac{\log_x 27 - \log_{x^2} 81}{2} = 1$$

$$\log_x (3 \times 9) - \frac{\log_{x^2} 9^2}{2} = 1$$

$$\frac{\log_x 3 + \log_x 9 - 2 \log_{x^2} 9}{2} = 1$$

$$\log_x 3 + \log_x 9 - \log_x 9 = 1$$

$$\log_x 3 = 1$$

$$\underline{x = 3}$$

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